

A Review of Microwave Wireless Techniques for Human Presence Detection and Classification

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(Invited Paper)

Abstract—Developments in microwave and millimeter-wave systems have enabled remote sensing techniques traditionally used in long-range applications to be employed in the relatively closer range applications of detection and classification of human presence and measurement of human properties. This paper reviews the techniques used in microwave remote sensing of humans and discusses prominent examples of experimental systems from the literature. Radar techniques have been developed for measuring the time-varying scattering returns from the various parts of the human body during motion, which are used for human presence detection and the classification of human activities. The effects of breathing and heartbeat are also detectable using Doppler radar techniques, which is applicable to medical applications as well as search-and-rescue and security. Recent developments in distributed radars have enabled the direct measurement of the angular velocity of moving people, and passive systems have also been developed which provide a complementary detection mode that can aid in detection and classification of humans.

Index Terms—Doppler radar, human classification, human presence detection, micro-Doppler, radar, radiometry, respiration.

I. INTRODUCTION

THERE has been a recent expansion in the development of microwave wireless systems for the remote detection and noncontact measurement of humans and human properties. Radar and radiometer technology, traditionally dominant in long-range applications, such as satellite remote sensing, radio astronomy, and military radar, is being adopted and developed for relatively shorter range terrestrial applications, including intruder detection, contraband imaging, and health diagnostics. Much of the technological development in these fields has been in the area of microwave and millimeter-wave systems in particular. Whereas phenomenology underpins the fundamental operation of such noncontact systems, it has largely been due to research and development at the system level that such human measurement techniques are now viable. For example, continuous-wave (CW) radar transceivers developed along with advanced time-frequency processing techniques have enabled micro-Doppler detection of humans, classification of human activity, and noncontact measurement of respiration and heart rate.

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Certain aspects particular to microwave systems have fueled the interest in remote sensing of humans. Microwave and millimeter-wave radiation propagates easily through most clothing materials, as well as obscurants, such as fog and smoke, allowing systems to measure properties of the human body directly, whether for the detection of human presence or biometric measurements. Microwave and millimeter-wave radiation also propagates through many building materials, such that through-wall radar is feasible for law-enforcement and search-and-rescue applications. The wavelengths are sufficiently small that the system aperture and front-end electronics can be made compact, good resolution (both spatial and temporal) can be achieved, and the continual improvements in microwave device technology have enabled lower cost, high-performance systems to be developed at frequencies extending through W-band.

This paper presents a review of microwave and millimeter-wave wireless techniques used to detect human presence and to classify human activities. Prominent examples of these approaches from the literature are reviewed; an exhaustive list of all the works in these areas is not intended; however, the works reviewed present a broad and representative overview of the current state of microwave and millimeter-wave remote measurements of humans. In Section II, the characteristics of some materials at microwave and millimeter-wave frequencies are reviewed. Following this, an overview of radar functionality and radar designs for human presence detection are given in Section III. Section IV reviews examples of radar applied to human presence detection and classification, focusing on Doppler radar signatures. Next, in Section V, a review of remote biometric measurements, focusing on respiration and heart rate, is given. Passive radiometer methods for human presence detection are discussed in Section VI. Finally, future directions are discussed in Section VII.

II. MEDIA CHARACTERISTICS AT MICROWAVE AND MILLIMETER-WAVE FREQUENCIES

In active and passive applications of remote sensing for human presence detection or classification, the radiated signal (whether actively transmitted and scattered off the person or intrinsically generated by thermal electromagnetic emission) propagates through a few different media. It is helpful, therefore, to review the propagation characteristics of some relevant materials at frequencies in the microwave and millimeter-wave region, as denoted in Table I, before describing the radar and radiometer applications in Section III.

TABLE I
IEEE STANDARD FREQUENCY BAND LETTER DESIGNATIONS

Band designation	L	S	C	X	K _u	K	K _a	V	W	mmw
Frequency (GHz)	1-2	2-4	4-8	8-12	12-18	18-26.5	26.5-40	40-75	75-110	110-300

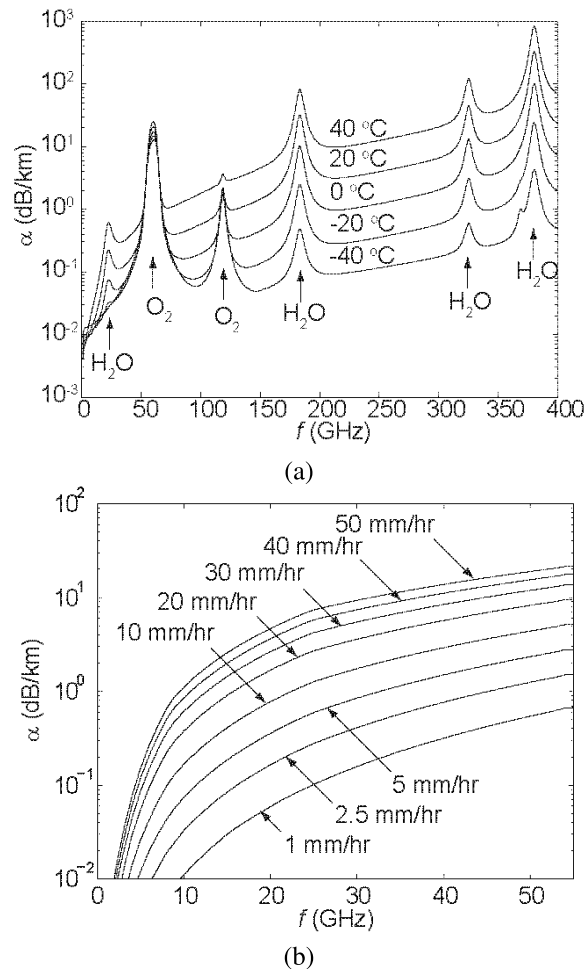


Fig. 1. (a) Atmospheric attenuation at sea level for various temperatures (from [1]). (b) Attenuation due to rainfall of various rates (from [1]).

In all cases, the radiation propagates through a small portion of the atmosphere. At lower microwave frequencies, around 10 GHz and below, the effects of the atmosphere are typically so minimal at the ranges of interest in the remote sensing of humans that they may be disregarded completely. However, at frequencies extending above 10 GHz into the hundreds of gigahertz, the effects of the atmosphere must be considered, sometimes even for short-range applications. Fig. 1 shows the attenuation of the atmosphere and that due to rainfall [1]. There are a number of constituents and properties of the atmosphere that contribute to the absorption of electromagnetic energy, although some of the most prominent properties are temperature, water vapor density, and the various resonances due to oxygen and water. Fig. 1(a) shows the trends as a function of temperature, where increasing temperature results in increasing attenuation. There is a general upward trend of the attenuation with frequency that is due to water vapor; this effect begins to plateau around the terahertz region.

TABLE II
MEASURED -3-dB ATTENUATION POINT OF COMMON MATERIALS
(ADAPTED FROM [3])

Fabric	-3 dB point
Wood (2.2 mm)	350 GHz
Linen (1.1 mm)	350 GHz
Leather (0.75 mm)	400 GHz
Denim (0.96 mm)	500 GHz
Naugahyde (0.65 mm)	700 GHz
Silk (0.36 mm)	1 THz
Nylon (0.19 mm)	1 THz
Rayon (0.15 mm)	>1 THz

Water absorption accounts for four principal resonances below 400 GHz, located approximately at 22, 183, 323, and 380 GHz. Oxygen accounts for two major resonances at approximately 60 and 118 GHz. There are clear operating windows in the spectrum between these resonances where long-range detection is possible, although some applications may make use of the increased attenuation at a resonance, such as 60 GHz, to minimize interference with other systems in the same band. Rainfall also affects attenuation, as shown in Fig. 1(b). Rain rates of 1–5 mm/h constitute light to moderate rain, while 10–30 mm/h is a heavy rain. A rain rate of 40–50 mm/h is an extreme downpour. It can be seen that the attenuation due to even moderate rain rates is almost negligible at low microwave frequencies, particularly at short ranges. In addition to rain, the attenuation through fog, smoke, and other obscurants has been measured [2], and is generally negligible at frequencies extending up to 100 GHz.

It is usually the case in microwave and millimeter-wave remote sensing of humans that the radiation propagates through clothing material; therefore, it is important to determine if any attenuation will occur. This is particularly true for passive systems that detect thermal emissions from the human body, since lossy materials will radiate thermal energy as well, which may interfere with the measurement of the intrinsic thermal emission. The propagation loss through a number of garment materials was characterized in [3], where all materials exhibited less than 3-dB loss through the frequencies of 300 GHz. Table II shows the measured 3-dB loss frequency for the set of materials measured. Attenuation through garment materials was also measured at 94, 326, 584, and 1024 GHz in [4]; in this paper, both parallel and perpendicular polarization were measured separately. Table III gives a summary of the measurements at 94 GHz, which are most relevant to the types of radar and radiometer sensors used in human presence detection and classification.

In search-and-rescue and security applications, the radiated signals often must propagate through building walls or other materials used in construction. The attenuation of various

TABLE III

MEASURED ATTENUATION (IN dB) OF COMMON GARMENT MATERIALS
(ADAPTED FROM [4])

Fabric	Attenuation at 94 GHz	
	Polarization	⊥ Polarization
Cotton shirt (0.3 mm)	0.2	0.1
Denim (0.64 mm)	0.7	0.7
Drapery (0.89 mm)	0.3	0.5
Leather (1.3 mm)	0.7	0.6
Sweater (2.1 mm)	0.4	0.4
Sweatshirt (2.1 mm)	0.3	0.2

TABLE IV

MEASURED ATTENUATION (IN dB) OF SELECTED BUILDING MATERIALS
(ADAPTED FROM [4])

Material	Attenuation at 94 GHz	
	Polarization	⊥ Polarization
Cardboard (3.9 mm)	1.2	1.3
Maple (12.7 mm)	10.8	7.1
Plywood (12.7 mm)	8.7	10.8
SPF (12.7 mm)	10.8	5.8
Concrete (11.1 mm)	9.8	10.5
Drywall (12.7 mm)	2.2	2.8
Glass (2.2 mm)	4.5	4.3
Brick (15.1 mm)	15.1	16.33
Cinder block (17.0 mm)	26.6	26.9

materials was measured in [5] at frequencies extending above 100 GHz. It is notable that at an 11-GHz measurement frequency, the total one-way attenuation for painted wood boards, plywood, glass, and various other materials was less than 7 dB. Concrete brick attenuation in [5] was measured at more than 20 dB. In [4], the attenuation of various building materials was measured at 94, 326, 584, and 1024 GHz in both parallel and perpendicular polarization. Table IV shows the measured attenuation at 94 GHz for a selection of the materials measured; a more complete table is available in the original work; however, Table IV summarizes the differences between materials at roughly the same thickness.

In general, microwave and millimeter-wave radiation propagates through clothing material with minimal attenuation. Thus, for most applications, the signals can be considered to be those either reflecting off the human body or generated by the body. Propagation through building materials, on the other hand, depends highly on the operating frequency and the type of material. Drywall and glass exhibit fairly low attenuation through 94 GHz; however, concrete and cinder block exhibit high attenuation at microwave and millimeter-wave frequencies. Wood materials exhibit low enough loss at microwave frequencies to enable through-wall sensing for most applications; however, at millimeter-wave frequencies, the attenuation approaches 10 dB, which can make sensing challenging.

III. RADAR SYSTEMS AND APPLICATIONS

A. Radar Overview

Radars are active remote sensing systems that illuminate an object or scene and process the signals that are scattered back to the receiving antenna. When illuminating an object that is moving, such as a walking person or the chest movements of a person breathing, a simplified version of the signal received at the antenna is given by [1]

$$s_{RX}(t) = \alpha e^{-j2\pi f \left[t + \frac{2}{c} \sum_n R_n(t) \right]}$$

where α is the amplitude, which is affected by the scattering properties of the human body and the propagation, f is the carrier frequency of the transmitted signal, c is the speed of light, and $R_n(t)$ is the time-varying range to the n th moving part of the person. Because human movements are temporally dynamic, the ranges to each body part change nonlinearly as a function of time, yielding time-varying Doppler frequency shifts.

For example, a simplified model of a walking person would take the arms as two scattering objects, the legs as two scattering objects, and the rest of the body as a single, larger scattering object. The signal reflected back from a walking person then contains five frequency shifts, four of which are temporally dynamic due to the movement as the arms and legs move forward and back, and one frequency, which is largely a static frequency shift, due to the bulk motion of the torso. In reality, the scattered signal is more complicated. However, this simple model captures the principal aspects involved in Doppler detection and classification of people.

Human respiration and heartbeat also generate time-varying signals with multiple frequency components. The chest moves forward and backward cyclically during the breathing cycle, and on top of this movement are smaller movements due to pressures generated by the heartbeating. Thus, one can measure both the frequency of respiration and the heart rate simultaneously if the frequencies in the return signal can be resolved.

It is these changes in phase imparting frequency shifts and their time-varying nature, the structure of which can be reduced to recognizable signatures in many cases, that are principally of interest in microwave and millimeter-wave remote measurements of human properties. Range and amplitude, used far less frequently in these applications, contain useful information in some cases. However, the dominant source of information is contained in the phase, and because of this, specific forms of radar hardware and signal processing have become preferred for remote human measurements.

The ability to resolve signals in frequency is dependent on the integration time; thus, it is important that a coherent waveform is used. CW radar systems, where a single tone is generated and received, are a popular choice due to its inherent simplicity. To implement a CW radar, without bandwidth or any temporal constraints on the waveform, a simple oscillator may be used as the transmit signal generator, and the receiver, after downconversion, need not have a fast digitizer. The main property is waveform length that is used for integration, and this is then determined in digital processing. While CW radar

simplicity in hardware and processing is a benefit, the inability of a CW radar to determine range can be a drawback in some applications. Frequency-modulated CW (FMCW) radars impart a modulation onto the CW tone, thereby allowing some range information to be calculated in addition to measuring the frequencies. Pulse Doppler radars are also used, where the CW signal is periodically amplitude modulated in an ON–OFF-keyed format, enabling higher peak power in the transmitted signal while maintaining a lower average power. While providing range, pulse Doppler radars also yield better sensitivity, since the peak power is greater. The drawback is the presence of range ambiguities resulting from the periodic “sampling” of the CW waveform.

Resolution in the frequency domain is dependent on the integration time. However, better resolution with longer integration time is only achieved if the signal is not varying in frequency over the integration time. Because many of the signals of interest generate time-varying Doppler shifts, the ability to resolve the separate frequency components is, therefore, limited. A popular way of representing of the scattered signals is in the time-frequency domain. The short-time Fourier transform (STFT) is the most commonly used time-frequency transform, which subdivides the received radar signal into shorter temporal segments using a windowing function $w(t)$, which are then processed in series through a standard Fourier transform; the resulting transforms are combined into a time series to form the time-frequency plot. The STFT is defined by [6]

$$\text{STFT}(t, f) = \int_{-\infty}^{\infty} s_{RX}(t')w(t' - t)e^{-j2\pi ft'} dt'.$$

The magnitude of the STFT is called the *spectrogram*. The design of the window function affects the resolution in both time and frequency, although inversely: implementing a short-time window yields a wide frequency window and vice versa; thus, the resolution can only be improved in one domain at the expense of resolution in the other. However, the field of time-frequency analysis is well developed, and many other time-frequency processing techniques have been developed to improve the ability to process signals in both domains [6]; a thorough review of these techniques is outside the scope of this paper.

B. Micro-Doppler

The term applied to the frequency sidebands generated by radar signals reflected off of secondary scattering centers in nonrigid bodies is *micro-Doppler*, and is detailed in [1] and [7]. When a radar signal is incident on a walking person, the largest return is typically produced by the torso, since it has the largest area, while the amplitudes of the signals scattered by the legs and arms are lower. However, the temporal variation of the frequencies from the arms and legs is distinct and periodic; these micro-Doppler signatures are used for detection and classification. Because most building materials simply attenuate radar signals, these features can also be discerned in through-wall scenarios if the signal levels are sufficient [9]; detection and classification through foliage have also been demonstrated [10], [11].

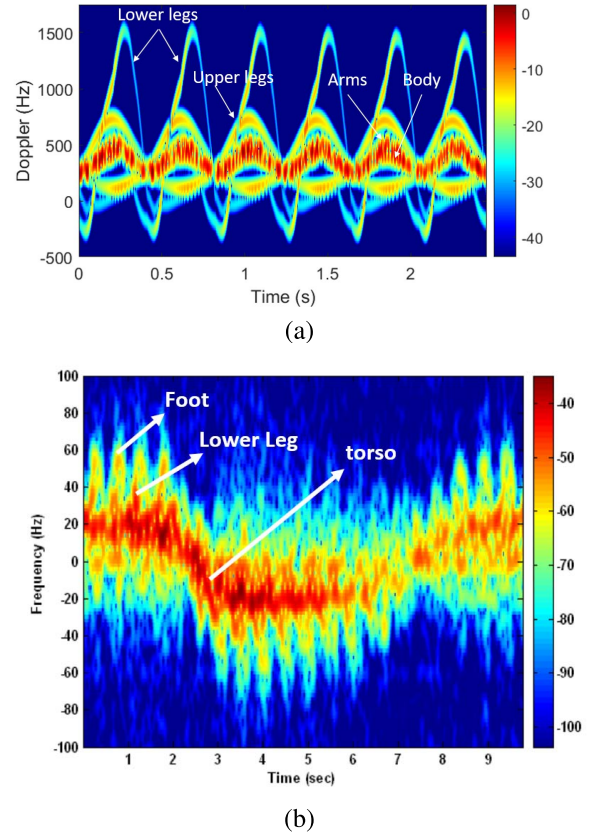


Fig. 2. (a) Simulated micro-Doppler signature of a person walking toward a 30-GHz radar at a range of 10 m. The signature is dominated by the frequency sidebands generated by the swinging motion of the legs, which have the highest velocity and, therefore, the greatest frequency shift. The periodic nature of the signature is also clear (generated with code from [7]). (b) Measured micro-Doppler signature of a walking person using a 2.4-GHz CW radar. The person walks toward the radar and turns around at 2.5 s, then walks away from the radar, turning around again at 7 s (from [8]).

The use of micro-Doppler analysis of moving people was pioneered by Chen [7] who presented a detailed model of the moving body based on work in [12] and provided MATLAB code for generating simulated micro-Doppler signals. Fig. 2(a) shows the simulated time-frequency signature of a person walking toward a 30-GHz radar at a starting distance of 10 m. The signatures from the swinging motion of the legs have the greatest frequency spread, and the cyclical nature of the motion of the legs and arms is apparent. The lower legs, moving with the highest velocity, produce the largest frequency shift, while the motion of the other moving parts produce less frequency shift but mostly have a greater signal amplitude. The periodic nature of the signature, being tied directly to the physical movement of the various body parts, is one of the factors enabling discrimination and classification of humans. Fig. 2(b) shows the measured spectrogram of a walking person using a 2.4-GHz CW radar [8]. The person walks toward the radar and turns around at 2.5 s, then walks away from the radar, turning around again at 7 s. The oscillatory pattern due to the leg motion is clear, as is the bulk frequency shift due to the torso. The ability to detect and process the minor signature variations, such as those due to the motion of the arms, constitutes one of the challenges of micro-Doppler classification.

Predicting and characterizing the minor movements of the human body, as well as its movement in three dimensions, are complicated endeavors. In [7], the simulated motion was only a walking gate in a linear direction. Others have investigated approaches to implementing more complex movements and predicting the resulting micro-Doppler signatures. For example, in [8] and [13], computer animation generated from a motion capture database of moving humans was used to predict micro-Doppler signatures, and in [14] and [15], an attempt was made to reconstruct a micro-Doppler signature using the sensors in a Microsoft Kinect as a motion capture device.

C. Interferometric Radar

A principal drawback of measuring a noncooperative moving person with a Doppler radar is the inherent reliance on the radial velocity of the person relative to the radar. If the person is walking tangentially to the broadside direction of the antenna, theoretically, no Doppler frequency shift is produced. Furthermore, other investigations into the use of micro-Doppler for classification have shown that the micro-Doppler signature is reduced in frequency significantly enough beyond a trajectory of about 60° relative to broadside to prevent sufficient features for classification [16]–[18]. It is important to be able to detect and classify people moving regardless of their trajectory relative to the radar, which would be enabled by a measurement of angular velocity in addition to the Doppler measurement of radial velocity.

One approach to measuring angular velocity was developed recently and utilizes an interferometric receiver composed of two antennas separated by a large number of wavelengths, the outputs of which are cross correlated [19]. As an object or person moves through the interference pattern generated by the interferometer, a voltage signal is produced whose frequency is proportional to the angular velocity of the object.

Assuming that the receivers connected at the outputs of the antennas are identical, the signals produced are proportional to

$$s_{RX1}(t) = Ae^{j2\pi f_c t}, \quad s_{RX2}(t) = Ae^{j2\pi f_c (t + \tau_g)}$$

where A is the amplitude of the received signal, $\tau_g = (D/c)\sin\theta$ is the geometric time delay of reception of the signal between the two antennas, and θ is the angle off broadside. The cross-correlated response is

$$r(\tau_g) = |A|^2 e^{j2\pi f_c \frac{D}{c} \sin\theta} = |A|^2 e^{j2\pi \frac{D}{\lambda} \sin(\omega t)}$$

where $\theta = \omega t$ and ω is the angular velocity. Near broadside ($\theta \ll 1$), the instantaneous frequency of the response is given by

$$f_s = \omega D / \lambda.$$

Compared with the Doppler frequency shift

$$f_d = 2v_r / \lambda$$

it can be seen that the two signals are mathematically almost identical. In the interferometric case, the frequency shift is proportional to the carrier frequency and the antenna baseline, whereas the Doppler frequency shift is proportional to the

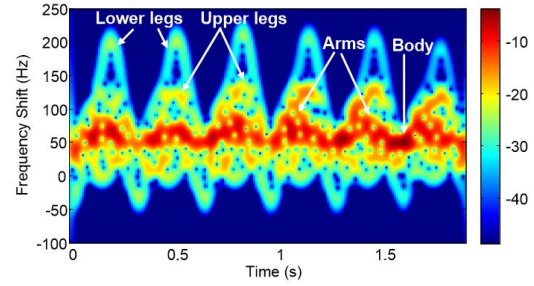


Fig. 3. Simulated interferometric microsignature of a person walking tangentially in front of a 30-GHz interferometric radar with a 40λ baseline at a range of 10 m (from [24], generated with a modified version of the code from [7]).

carrier frequency. In the same manner that the sign of the Doppler frequency indicates whether the person is approaching or receding from the radar, the sign of the interferometric frequency shift indicates whether the person is moving clockwise (right to left, indicated by positive frequencies) or counter-clockwise (left to right, indicated by negative frequencies). Thus, combining the interferometric method with traditional Doppler radar provides a method of directly measuring the motion of a person regardless of their trajectory. Whereas multistatic radars can perform similar tasks by measuring a person from various sensor locations and viewpoints [20], [21], the interferometer can provide motion measurements from a single sensor position.

While mathematically nearly identical, one significant difference between the Doppler and interferometric approaches is the nonlinear nature of the interferometric correlation processing. Thus, scattered signals with multiple frequency components will generate intermodulation products in the correlation process, which can make reproducing a micromotion signature similar to that seen in micro-Doppler challenging. However, there are approaches to reducing the effects of the nonlinear processing to recover the signal, as described in [22] and [23].

Fig. 3 shows the simulated response of a person walking tangentially to an interferometric radar at a distance of 10 m when the nonlinear distortion of the correlation process is mitigated. It can be seen that the arms and legs produce a very similar microsignature to that of the micro-Doppler radar; this is not unexpected, of course, since the signatures still derive from the same physical motion of the person, and the mathematical representation of the two frequency shifts is nearly identical. Because the interferometer can produce signatures representative of the same physical information as the Doppler radar, but can do so when the person is moving tangentially to the radar, a full trajectory picture can be produced regardless of the relative motion of the person.

IV. HUMAN DETECTION AND CLASSIFICATION

The utility of the human micro-Doppler signature is in the analysis of the time-frequency representation to detect and classify human activities [17], [18], [25]–[43]. Features in the signature, such as the periodicity of the leg swing component, the frequency spread of the signature, and the

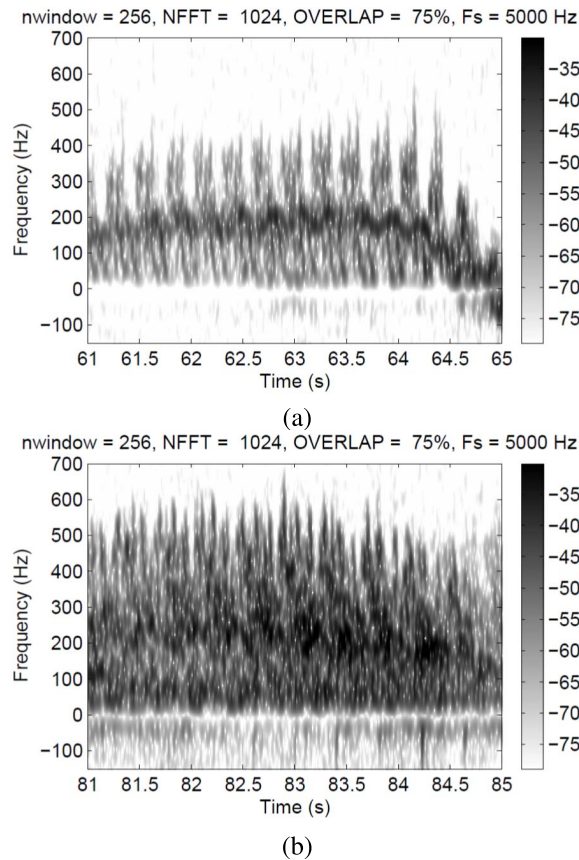


Fig. 4. Measured micro-Doppler of (a) dog and (b) goat measured at X-band. The differences due to the quadrupedal nature of the animal compared with the bipedal micro-Doppler signature of a person are apparent (from [17]).

velocity of the bulk motion, among others, are often particular to human motion, and to specific human motions in some cases. Many works focus on the extraction of appropriate features to perform robust classification. In the following, the results of a few prominent examples of classifiers from the literature are reviewed.

In [31], micro-Doppler signatures of walking people were compared with that of a horse and rider, showing the differences in the oscillatory pattern of the leg swing signature. In [17], the signatures of people, as well as dogs, goats, deer, and birds, were measured and analyzed with an X-band CW Doppler radar. The micro-Doppler signature of a dog and goat is shown in Fig. 4. It can be seen that the signature due to the leg motion, which generates the large frequency shift, is quite different than that of a person, where a single tone is clear; with a quadrupedal animal, multiple tones are present, creating a more complex, yet still oscillatory signal.

Classification algorithms are used to differentiate between return signals from different classes, which, for instance, may be the set of [human, animal, vehicle] or may be a set of activities, such as [walking, running, crawling]. Classification algorithms operate on a set of features that are present in the signature; these are identifying aspects that vary with the class, and thus can be used to differentiate between classes. Often, a large set of features is required to implement a robust classifier, although in some instances, a few or even one feature may be sufficient.

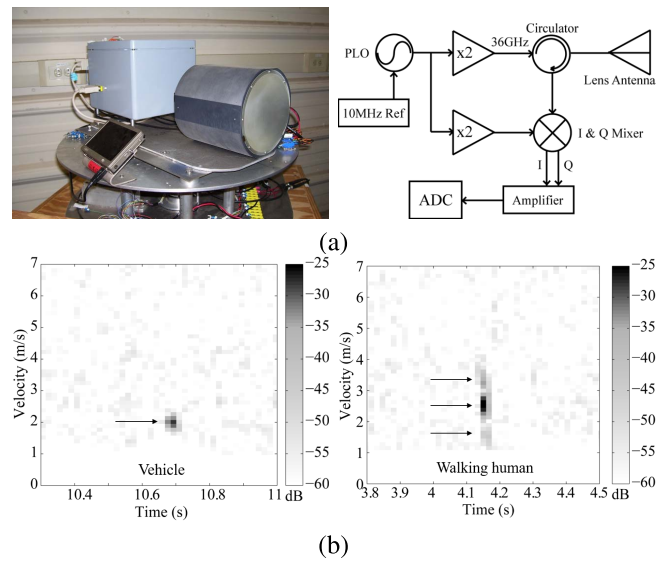


Fig. 5. 36-GHz CW Doppler radar for human presence detection and classification from a moving platform. (a) Radar system and block diagram. (b) Measured response of a vehicle (left) and a walking person (right) moving at approximately the same velocity. The multiple frequencies due to the various scattering centers on the person are easily distinguishable from the single scatterer of the vehicle (from [26]).

A. Examples: Human Detection

A simple and direct approach to classifying humans and vehicles that focused on the presence of micro-Doppler or lack thereof was presented in [26], where a naive Bayesian classifier was implemented. In this paper, a 36-GHz CW Doppler radar was implemented on a rotating platter atop a moving platform (see Fig. 5), and thus did not have the benefit of long integration time. The only feature used in this classifier was the number of frequency components in the STFT of the return signal: due to micro-Doppler, a person is likely to generate multiple frequency components whereas a car will not. Fig. 5(b) shows the measured responses of a vehicle and a person from the rotating radar, where the multiple frequencies in the signature of the person are apparent. A probability of detection of > 0.9 with a false alarm rate of ≈ 0.01 was achieved with 4–6 radar dwells (rotations).

Another classifier to differentiate between humans and other nonhumans was based on the measured data from which an X-band multifrequency CW radar with 16-dBm transmit power was reported in [17], where a classifier based on a support vector machine (SVM) was implemented. A large set totaling 80 features was evaluated, and a test called the Fisher score was used to determine the 33 features with highest discriminatory power. The result was a probability of detection of 0.942 with a false alarm rate of 0.116 using a staring radar approach. A Gaussian mixture model was also implemented in [17] using the same database, which yielded a probability of detection of 0.936 with a false alarm rate of 0.093.

In [27], an SVM was implemented to discriminate between humans, dogs, bicycles, and cars. It operated on the four features of limb motion frequency, target stride, Doppler frequency spread (bandwidth), and the difference in the strength

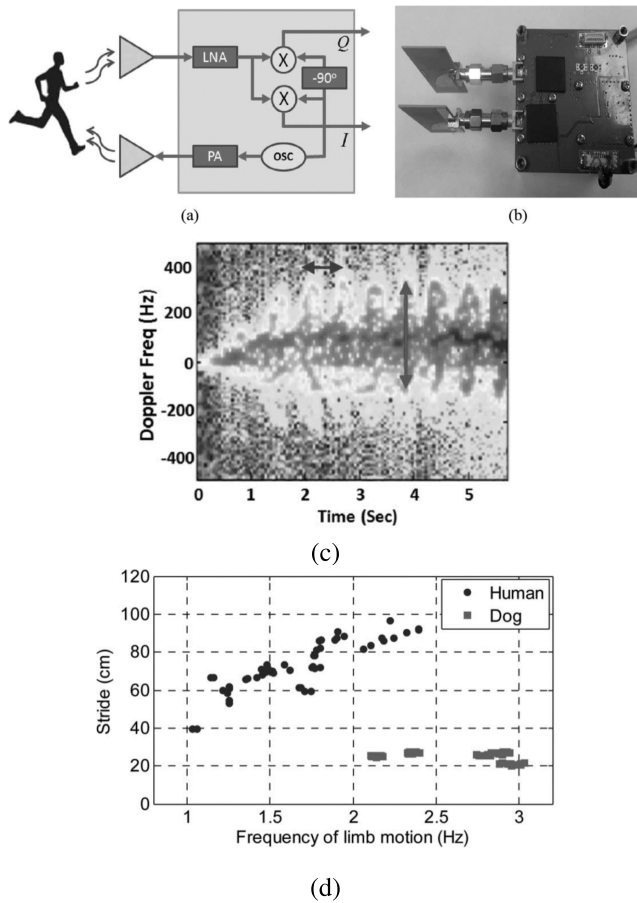


Fig. 6. (a) 7.25-GHz Doppler radar schematic and (b) image of the radar. (c) Example spectrogram of a walking person indicating where the features are measured. (d) Feature distribution for humans and dogs (from [27]).

of the signals corresponding to the different body parts. The radar, shown in Fig. 6, operated at 7.25 GHz with an output power of 15 dBm and utilized a direct-conversion I/Q receiver. Two of the features utilized in the classifier, the stride and Doppler bandwidth, are shown in Fig. 6(c), and the distribution for the two classes of human and dog is shown in Fig. 6(d). The accuracy of the classifier was 96%, and it correctly classified humans, dogs, bicycles, and cars at the rates of 96%, 90%, 76%, and 79%. Using the same radar, a deep convolutional neural network (DCNN) was implemented in [44], where discrimination of humans from vehicles, horses, and dogs was achieved with 97.6% accuracy, and classification of four human activities (running, crawling, boxing while standing, and boxing while moving forward) was achieved with an accuracy of 90.3%. A DCNN was also applied to 7.25-GHz measurements of swimming activities in [45], where a maximum classification accuracy of 80.3% was achieved.

B. Examples: Human Activity Classification

As shown in Fig. 7, a classifier based on an SVM was implemented in [18] to differentiate between different types of human activities, which were measured with a 2.4-GHz radar. The features used in the classifier were the Doppler frequency

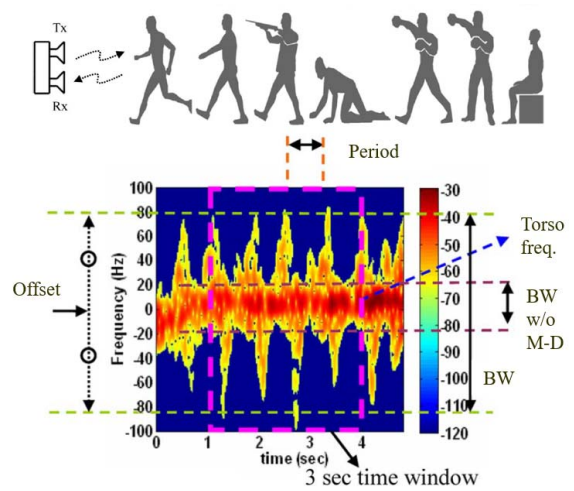


Fig. 7. Activities that were classified using an SVM operating on the various features of micro-Doppler signatures (from [18]).

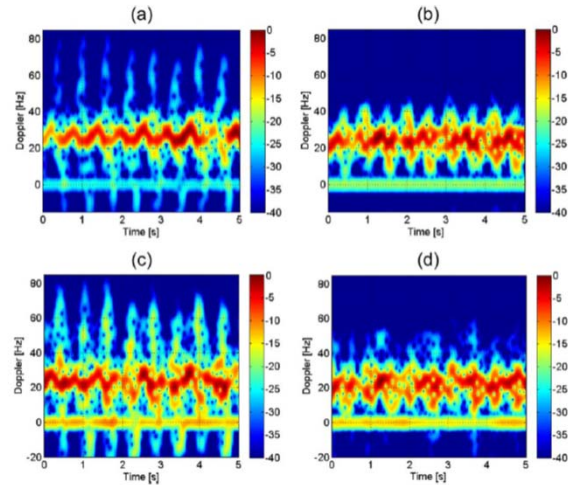


Fig. 8. Spectrograms of a person walking toward one node in a multistatic array. (a) Monostatic return of an unarmed person. (b) Monostatic return of an armed person. (c) Bistatic return from an unarmed person. (d) Bistatic return from an armed person (from [32]).

from the torso, the total frequency spread of the signature (total bandwidth), the total offset frequency, the bandwidth of the bulk signature (ignoring the micro-Doppler from the arms and legs), the standard deviation of the Doppler signal strength, and the period of the limb motion. Based on training on these features, the SVM was able to classify the seven different activities shown in Fig. 7 with an accuracy above 90%.

In [30], empirical mode decomposition was used to determine the frequency content of the return signal. An SVM was implemented using a calculated set of features extracted from a training set to classify the signal into a set of motions, including breathing, swinging arms, picking up an object, and standing from a crouching position. Two radars were tested, an S-band radar and a W-band radar; the S-band radar yielded an overall accuracy of nearly 75% while the W-band radar yielded an accuracy of 80%–90% depending on the motion.

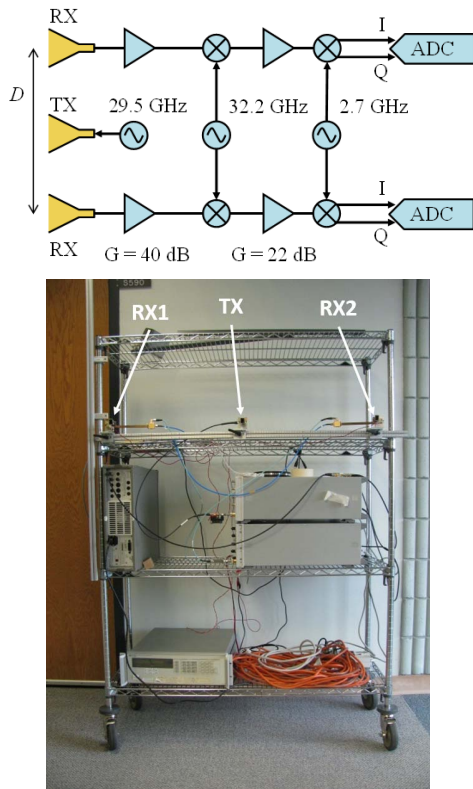


Fig. 9. Block diagram and image of a 29.5-GHz interferometric radar for measuring the angular velocity of moving people (from [24]).

A multistatic 2.4-GHz radar was used in [32] to classify between armed and unarmed personnel moving at various trajectories. The radar used a pulse Doppler waveform with 0.6- μ s pulse lengths with a 45-MHz linear frequency modulation and a transmit power of 200 mW. Fig. 8 shows the measured results of an unarmed and armed person in monostatic and bistatic measurements. Singular value decomposition was used to extract a single feature that was then used in a Naive Bayesian classifier, which achieved a classification accuracy of 97.2%.

C. Examples: Interferometric Radar

The interferometric angular velocity measurement technique was first demonstrated in [24] where a 29.5-GHz interferometric CW radar was implemented. The radar and its block diagram are shown in Fig. 9. The receivers were separated by 40λ , and a person walking tangentially back and forth in front of the radar was measured. The results showed that the appropriate frequency shift due to the angular velocity was achieved, and also that the directionality was indicated by the sign of the frequency: positive frequencies indicated clockwise movement (left to right) and negative frequencies indicated counterclockwise movement. The angular velocity measurement technique was combined with a Doppler measurement in [46] and [47], demonstrating that multidimensional trajectory information could be measured instantaneously without tracking. Experimental results are shown in Figs. 10 and 11, where a person walked in three

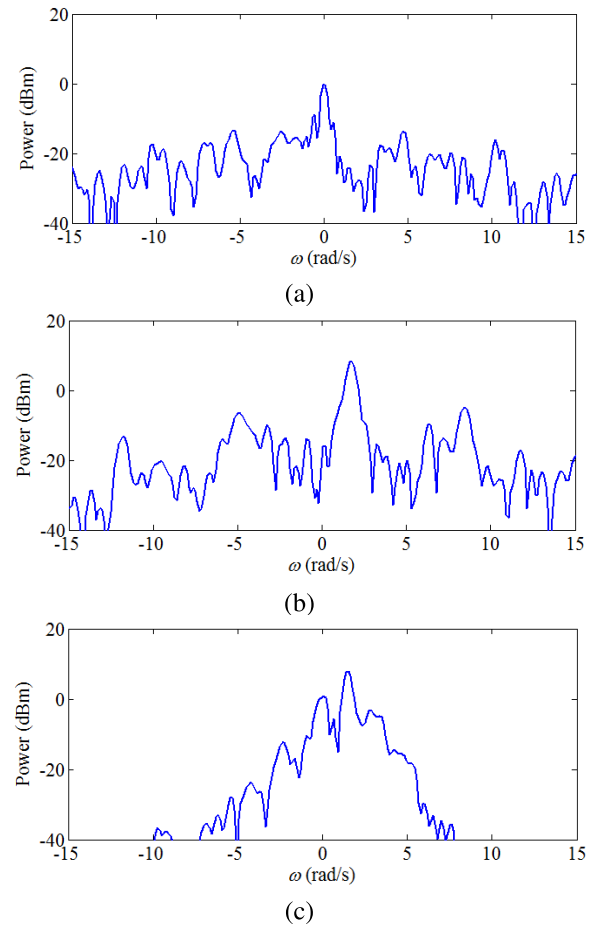


Fig. 10. Response of a 29.5-GHz interferometric radar with $D = 40\lambda$ to a walking person holding a corner reflector at relative angles of (a) 0° , (b) 45° , and (c) 90° . As the trajectory becomes angular, the frequency shift and angular velocity increase (from [46]).

different trajectories toward a joint interferometric-Doppler radar. The radar consisted of a 29.5-GHz transmitter and two receivers separated by 40λ . The receiver outputs were cross correlated, yielding the interferometric signal, and one receiver output was also processed directly, yielding the Doppler signal; therefore, the measurements were conducted simultaneously. Fig. 10 shows the interferometric signals from when the person walked at angles of 90° , 45° , and 0° relative to the radar, and Fig. 11 shows the Doppler signals for the same measurements. At a trajectory of 90° (radial), the interferometric frequency shift is zero while the Doppler shift is at its maximum; at a trajectory of 0° , the opposite is the case; when the trajectory is in the middle, both modes register a frequency shift. Thus, the interferometric measurement provides a complementary measurement to the Doppler radar.

The nonlinear processing used in the cross correlation can cause the micromotion signature of the interferometric radar to be indistinguishable due to intermodulation of the multiple frequency components present in the radar return. In [22], two methods were introduced to reduce the distortion and recover the underlying micromotion signature, one that used long wavelengths and one that used short pulses. In either case, upon sufficient mitigation, the underlying signature was shown in [46] to match the micro-Doppler signature almost

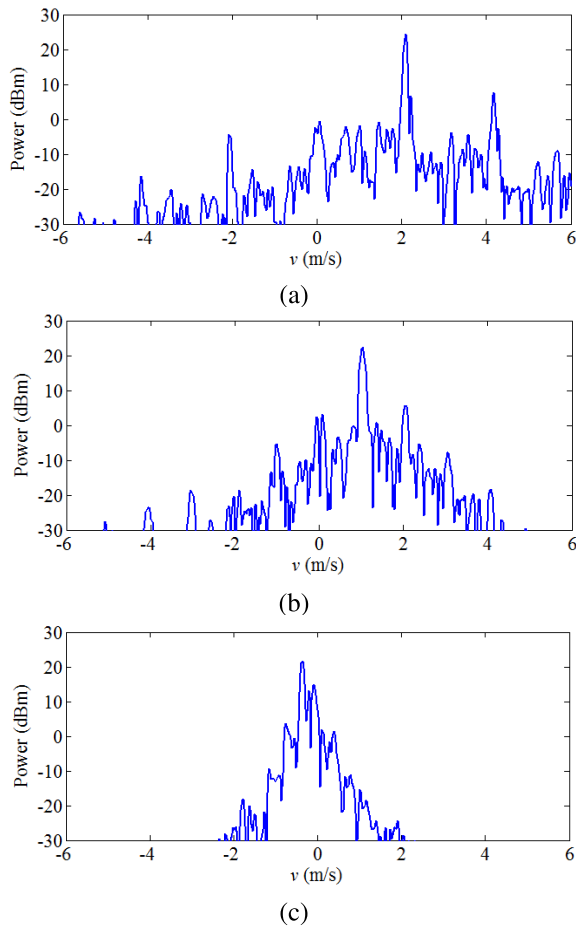


Fig. 11. Response of a 29.5-GHz Doppler radar to a walking person holding a corner reflector at relative angles of (a) 0° , (b) 45° , and (c) 90° . As the trajectory becomes angular, the frequency shift and radial velocity decrease (from [46]).

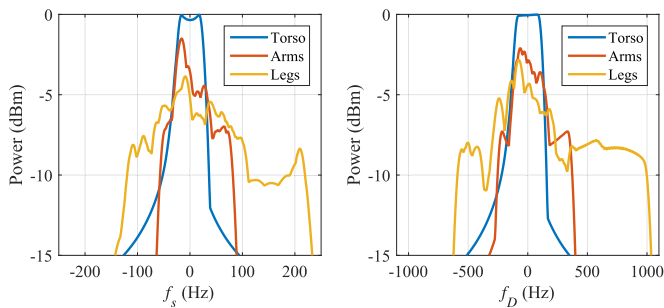


Fig. 12. Peak signal responses of a 30-GHz, 100λ interferometric radar viewing a person walking tangential to the radar (left) and a 30-GHz Doppler radar viewing a person walking toward the radar (right). The micromotion signatures show nearly identical spectra (from [46]).

identically (see Fig. 12). Thus, the same types of classification processing that is performed on micro-Doppler signatures may be applied to interferometric signatures as well, enabling human classification regardless of the trajectory relative to the radar.

V. MICROWAVE REMOTE BIOMETRIC MEASUREMENT

Remote measurements of biometric properties of the human body focus largely on measuring the time-varying phase of the return signal, which is modulated due to movements of the

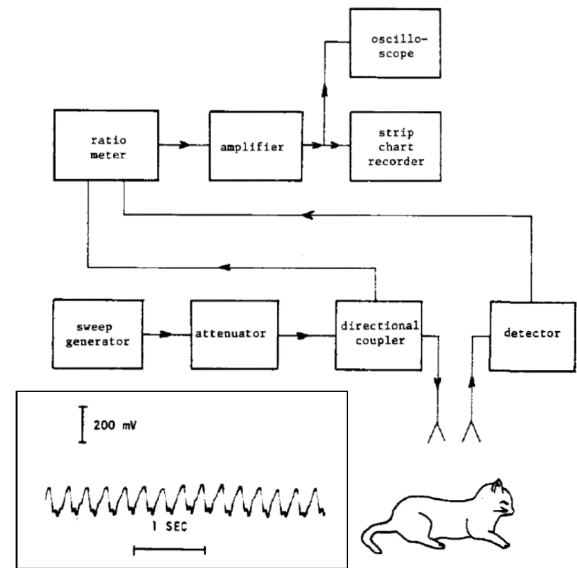


Fig. 13. One of the first microwave remote respiration measurements using an X-band radar. Inset: measured respiration of a rabbit (although the cartoon shows a cat) (from [48]).

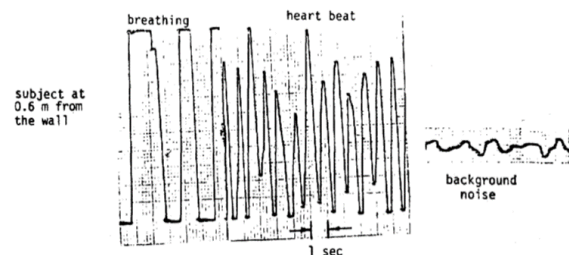


Fig. 14. X-band measurement of respiration and heartbeat through a cinder block wall (from [49]).

chest [48]–[69]. An excellent review of microwave radars for remote health monitoring is given in [70]. There is a direct utility in medical settings for noncontact biometrics, because it is more comfortable for the patient, but remote biometric measurements are also applicable to search and rescue, where life signals can be detected through building materials, whether standing walls or collapsed rubble in disaster scenarios.

Respiration induces the largest movements, while cardiac actions induce smaller movements that impart higher frequency modulations; the displacement due to the heartbeat is on the order of only 0.6 mm [71], and thus, high frequencies may be beneficial to ensure that larger phase shifts are imparted by the heartbeat motion. The challenge is then differentiating the respiration and heartbeat signals; additional processing may also allow measurement of heart arrhythmia, where the frequency of the heartbeat changes over time.

One of the first demonstrations of respiration measurements with microwave radar was presented in [48], where a 10-GHz CW radar with an output power of 10 mW was implemented and measured the respiration-induced movements of a person and a rabbit; Fig. 13 shows a diagram of the setup and measurements of the respiration of a rabbit. Measurements of the respiration of a person were also conducted, and even though the distance in this case was only 30 cm, the

measurements proved the feasibility of the remote microwave measurement of respiration.

Soon thereafter, researchers developed methods of measuring the heartbeat using microwave radiation in a contact applicator [72], [73]; however, these used physical applicators in contact with the chest wall. In [49], a remote life sign detection radar was demonstrated. The X-band radar used a transmit power of 4.5 mW and was able to detect respiration signals as well as heartbeat signals at distances of 30 m, or through a cinder block wall at a distance of 3 m. Fig. 14 shows the measurement of respiration and heartbeat through a cinder block wall.

More recently, researchers have focused on developing radars for medical applications of noncontact respiration and heartbeat measurements [74]–[78]. A Ka-band heartbeat detection radar was demonstrated in [58] for measuring a person at various viewing angles. Fig. 15 shows a block diagram of the experimental setup and measurements from the front and back of a person. The heartbeat signal shows up more clearly when measured from the back.

One of the challenges of using CW Doppler radars in this context is the effect of range nulling, where the signals cancel out with distance every quarter wavelength from the radar, since the return signal and the local oscillator are then 180° apart. This range can be on the order of centimeters or millimeters, depending on the frequency, and can be challenging to predict in noncontact measurement scenarios. Using a quadrature demodulator circumvents the range nulling problem, since it produces an in-phase component and a quadrature component that are 90° out of phase; thus, when range nulling occurs for one component, the other will still be present. This ensures that at least one signal is always available from which to extract the measurement.

In [50], a quadrature demodulator was implemented on two integrated radars operating at 1.6 and 2.4 GHz. The transmit power for each system was under 10 mW, and measured respiration and heart rate signal at a distance of 50 cm. The accuracy achieved for heart rate detection was above 80%.

A Ka-band multitone CW radar was shown in [59]. The authors chose a Ka-band signal to achieve better sensitivity in detecting small movements of the chest. A two-tone signal was implemented to eliminate range nulling in this case. This paper implemented a very low-power transmitter, using only 12.5 μ W, and achieved a detection accuracy of 100% at a distance of 0.5 m and an accuracy above 80% at a distance of 1.5 m using a low-gain single patch antenna. Replacing this with a high-gain array, the accuracy remained above 80% to a distance of 2 m.

An integrated CW Doppler radar with a quadrature receiver operating at 5.8 GHz using 7.5-dBm transmit power was demonstrated in [67] to measure chest movement due to respiration and heartbeat at a distance of 0.5 m. The quadrature receiver was implemented to improve the accuracy with which heart rate variability can be measured. Compared with electrocardiogram, the radar was able to measure the heart rate with a maximum relative error of 4.84%.

Laboratory instruments commonly used for communications testing were used for vital sign detection in [57]. A vector

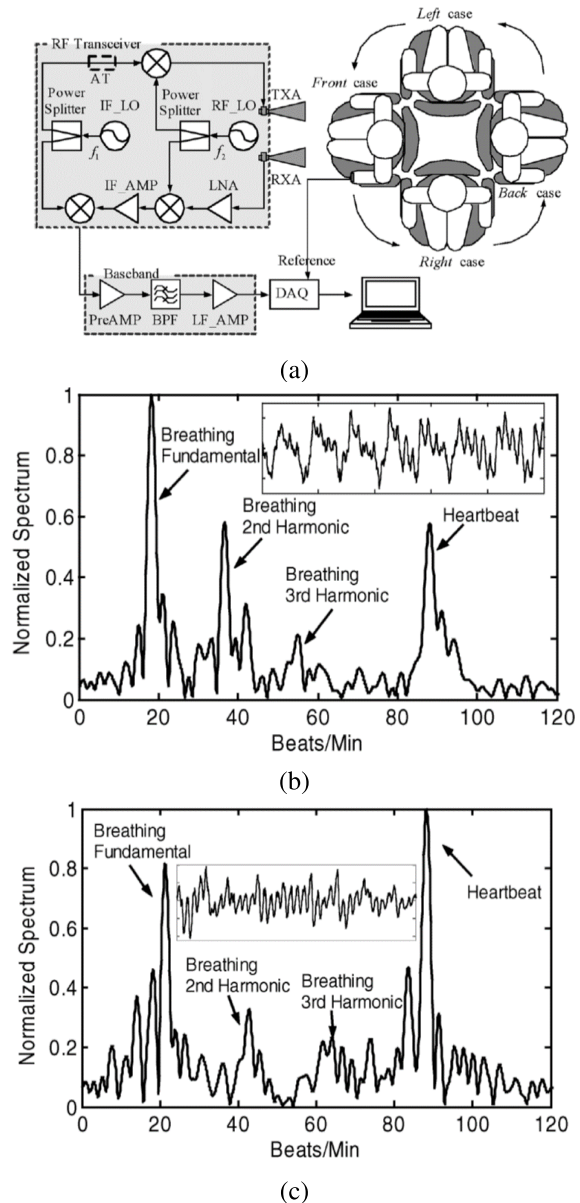


Fig. 15. Ka-band measurement of heartbeat at various viewing angles. (a) Experimental setup. (b) Measurement from the front of a person. Inset: time-domain signal. (c) Measurement from the back. Inset: time-domain signal. The heartbeat signal is more prominent when measured from the back (from [58]).

signal generator was used as a transmitter, a spectrum analyzer as the receiver, and a vector signal analyzer was used as a digital I/Q demodulator. Operating at distances up to 2.5 m, detection accuracies above 80% were achieved with a 0-dBm transmit power.

In [69], a multimode radar was developed to detect and measure life activity, including not only respiration and heart rate, but also micro-Doppler when the person moved. An overview of the 5.8-GHz FMCW radar is shown in Fig. 16, and remote measurements of the respiration and heartbeat of a sitting person as well as the micro-Doppler signal when a person begins to move are shown in Fig. 17. This system demonstrates the utility of combining various remote human measurement methods; in this case, life signs were used to

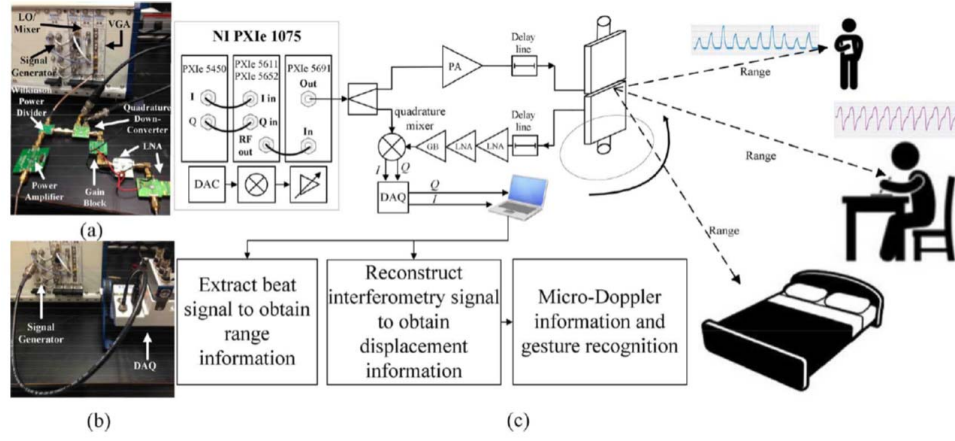


Fig. 16. 5.8-GHz life detection system: (a) hardware, (b) synchronization between the signal generator and DAQ, (c) block diagram of the radar system and the experimental setup (from [69]).

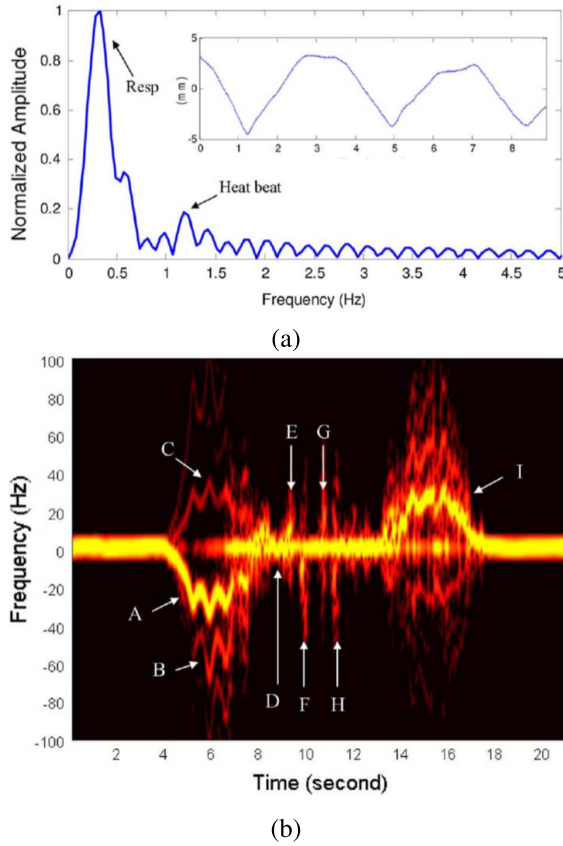


Fig. 17. 5.8-GHz life detection system. (a) Respiration and heartbeat measurement. Inset: time-domain signal. (b) Micro-Doppler measurement of a walking person. A–C indicate parts where the person is walking away from the radar. D–H indicate a pushing movement with the arms. I indicates when the person walks toward the radar (from [69]).

detect the presence of stationary people as well as walking people.

VI. RADIOMETER SYSTEMS AND APPLICATIONS

A. Radiometry Overview

Microwave and millimeter-wave radiometers are passive systems that detect and measure thermally generated

radiation [1]. The radiation signal of interest may come directly from the human body, or it may be background environmental radiation that is reflected off some object on the human body. In either case, the radiation in the microwave and millimeter-waveband generated by the human body is on the order of pico- or femtowatts; thus, radiometers must have very high gain to detect such radiation. Furthermore, the radiation is noiselike, so that sensitivity is crucial, since a matched-filter style of processing that aids in radar sensitivity cannot be used in with unknown noiselike signals.

Radiometers measure the *spectral brightness* of an object. The spectral brightness of a perfectly emitting material, called a blackbody, is given by Planck's blackbody law

$$B_f = \frac{2hf^3}{c^2} \frac{1}{e^{hf/kT} - 1}$$

where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant, f is the frequency, c is the speed of light, $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, and T is the temperature of the object. In the microwave and millimeter-wave region, it is generally true that $hf \ll kT$, and then, Planck's law reduces to the Rayleigh–Jeans Law

$$B_f = \frac{2f^2kT}{c^2}$$

where the spectral brightness is directly proportional to the physical temperature of the blackbody. For real materials, the spectral brightness is proportional to the temperature modified by its emissivity e , which is a ratio between zero and one, with the latter being a perfect blackbody. Thus, the brightness is given by

$$B_f = e \frac{2f^2kT}{c^2}.$$

Since the spectral brightness is measured in W/(m² Hz str), integrating the brightness over the receiver bandwidth, the receiving antenna aperture area A_r , and the angle subtended by the object Ω yields the power thermally radiated by the object. The power received by the radiometer antenna is thus

$$P = \int_f \int_{A_r} \int_{\Omega} B_f(\theta, \phi) \cos \theta d\Omega dA df.$$

TABLE V
EFFECTIVE EMISSIVITY AT NORMAL INCIDENCE OF COMMON MATERIALS
(ADAPTED FROM [80])

Material	44 GHz	94 GHz	140 GHz
Bare metal	0.01	0.04	0.06
Painted metal	0.03	0.10	0.12
Painted metal under canvas	0.18	0.24	0.30
Painted metal under camouflage	0.22	0.39	0.46
Dry gravel	0.88	0.92	0.96
Dry asphalt	0.89	0.91	0.95
Dry concrete	0.86	0.91	0.95
Smooth water	0.47	0.59	0.66
Rough or hard-packed dirt	1.00	1.00	1.00

A voltage signal proportional to the received power can be obtained in a few ways. The most common is by using a total power radiometer, which utilizes a square-law detector to produce a voltage signal that is proportional to the power at the antenna. Another method is a correlation receiver that uses two separate receivers and cross correlates (multiplies and integrates) the outputs of the two, thereby producing a voltage signal also proportional to the received power. The latter receiver has the benefit of a sensitivity improvement of $\sqrt{2}$, since the noise in the two receivers is uncorrelated.

Radiometers have uses in medical measurements of human body temperature [79], particularly since microwave and millimeter-wave sensors detect emissions from beneath the skin [1] whereas infrared sensors detect only the radiation of the surface of the skin. For the detection and classification of human presence, radiometers are utilized less frequently than radars, although they have benefits, including the ability to detect the thermal signature of a person even when they are not moving.

Objects at the same physical temperature can still be distinguished using a radiometer if the emissivity of the objects is sufficiently different. For instance, the emissivity of human skin at millimeter-wave frequencies is 0.5–0.7 [1], while that of unpainted metals is 0.01–0.06. Thus, even a metal object held on the body at the same temperature will register a lower power. The emissivity of various materials is given in Table V.

B. Passive Human Presence Detection

The simplicity of radiometers and their ability to respond directly to the temperature of an object in the antenna field of view makes them ideal for simple human presence detection tasks. A total power radiometer operating from 11–12 GHz was implemented in [81] to detect the presence of a person in front of an educational display. While no classification was implemented, low-cost components totaling in the hundreds of dollars were used to implement the radiometer, demonstrating that for simple tasks, the cost of implementing a fully passive microwave human presence detection system can be fairly low.

In [82], a 48-GHz radiometer was demonstrated to detect pedestrians and bicyclists. Directed slightly downward toward the road, the radiometer registered a simple voltage increases when a person or bicyclist moved past the antenna beam.

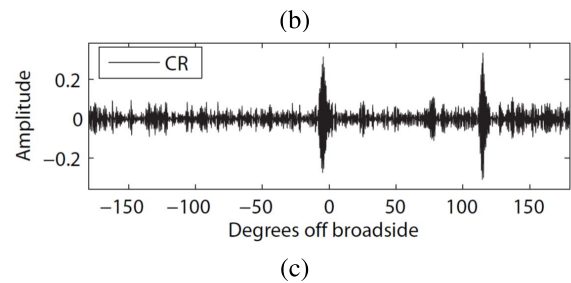
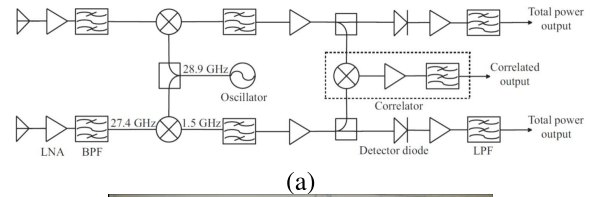


Fig. 18. 27.4-GHz passive correlation radiometer for detecting humans from a moving platform. (a) Radiometer block diagram. (b) Image of the radiometer atop the mobile robotic platform. (c) Example radiometer output showing a detection of a person near 0° and a false alarm near 120° (from [83]).

When a vehicle passed through the beam, the voltage initially increased when the front of the car is in the beam, and then decreased below the ambient level as the top of car, which reflected the cold sky, passed through the antenna beam. Thus, this paper indicates a simple set of features by which humans and vehicles could be classified based only on the voltage peaks and dips.

Detecting the thermal emissions from a person in more complex situations, such as from a moving vehicle, requires more sophistication. The concept of using a correlation radiometer was introduced in [83] and [84], where a two-channel radiometer operating at a center frequency of 27.4 GHz and a bandwidth of 500 MHz was developed to detect the presence of stationary people from atop a moving platform with a rotating antenna array. The correlation radiometer ideally responds only to incident radiation that is coherent between the two receivers; thus, any signals that are uncorrelated should be filtered out. Thus, any sources that are not viewed by both antennas should be removed from the signature. Fig. 18 shows the block diagram of the correlation radiometer along with an example of the correlated signal response over one full

rotation. The response near 0° is a person, and the response near 120° is a large metal background object at the same range as the person.

In [85], the correlation signal was combined with the total power responses of each channel. These signals were used as features in a Naive Bayesian classifier, which was developed to detect stationary people among a cluttered environment. A detection rate of 0.85% was achieved using a passive ranging method, while if the range to the object was known the detection rate increased to 92%.

VII. FUTURE DIRECTIONS

The feasibility of applying microwave remote sensing techniques to the detection and classification of human presence has been demonstrated through numerous experimental investigations. Active radar systems, distributed radars, and passive systems can all provide capabilities for applications, including security, medical diagnostics, and search-and-rescue individually, and one future direction is the combination of multiple functions for robust classification performance, one example of which was shown in [69]. Combinations of microwave and millimeter-wave remote sensing applications with other modalities, such as optical or infrared, can provide robust capabilities by integrating measurements in separate domains. No single detection method or system can perform perfectly in all conditions, and in particular for applications, such as security and search-and-rescue where missed detections are very costly, the combination of multiple sensors holds significant potential. Sensor fusion and data fusion represent an important area of research for future microwave and millimeter-wave remote sensors.

For microwave and millimeter-wave hardware, an important future direction is the integration of radars and radiometers into compact, low-power, low-cost packages. Particularly important for broad adoptions of these methods, such integrated systems also enable more seamless integration into the environment, and easier approaches for physical implementations of multimode sensor suites. Applications, such as hand-motion recognition and classification, are becoming of greater interest [86], [87] and will require integrated solutions. Integration into board-level and chip-level systems is currently an ongoing area of research and should only increase in activity going forward.

Another future aspect is the ability to coordinate separate wireless systems or chipsets to operate as a distributed system. At minimal levels of coordination, detection and classification can be improved through multiple simultaneous measurements of different aspect angles. At greater levels of coordination, increased range with noncoherent radar processing can improve detection capabilities. One area of active research is in the coherent combination of distributed wireless sensors, where a group of systems coherently operates as a distributed phased array. Such coordination would enable significantly longer detection ranges, as well as more sophisticated techniques, such as the interferometric measurement technique described earlier.

VIII. CONCLUSION

Remote detection and measurement of humans using microwave and millimeter-wave wireless systems enable new capabilities for security, search-and-rescue, and medical applications, among others. Various techniques have been reviewed in this paper, and their experimental demonstrations show the feasibility of such techniques. Classification of human activity for improved security and search-and-rescue is viable, and new developments in angular velocity measurements point to the potential for continually improving detection and classification. Passive systems can also provide a beneficial complementary measurement to the more commonly used active measurement techniques. There is strong interest in microwave remote sensing for human presence detection and classification in numerous fields, and there exists good potential for future research and development.

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